

HW1:

求下列JAVA程式的執行結果。

```
public class A {  
  
    public static void main(String[] args) {  
        C c = new C();  
        c.foo();  
        c.bar();  
    }  
  
    public void foo() {  
        System.out.println("foo from A");  
    }  
  
    public void bar() {  
        System.out.println("bar from A");  
    }  
}  
  
class B extends A {  
    @Override  
    public void foo() {  
        System.out.println("foo from B");  
    }  
}  
  
class C extends B {  
    @Override  
    public void bar() {  
        System.out.println("bar from C");  
    }  
}
```

Ans :

foo from B

bar from C

HW2:

Fix the following program, make it OO.

```
import java.util.ArrayList;
```

```
public class EmployeeCensus {  
    private ArrayList<Employee> employees = new ArrayList<>();
```

```
    public void addEmployee(Employee employee) {  
        employees.add(employee);  
    }
```

```
    public void removeEmployee(Employee employee) {  
        employees.remove(employee);  
    }  
}
```

```
class Employee {  
    // not important  
}
```

HW3:

求下列JAVA程式的執行結果。

```
public class X {  
    public static void main(String[] args) {  
        X x = new X();  
        Y y = new Y();  
        x.doSomething();  
        y.doSomething();  
    }  
  
    public void doSomething(){  
        this.print();  
    }  
  
    public void print(){  
        System.out.println("XXX");  
    }  
}  
  
class Y extends X{  
    @Override  
    public void print(){  
        System.out.println("YYY");  
    }  
}
```

Ans:

XXX

YYY

HW4:

The following code is Java. There are some compilation errors in the code.

How can you fix it to pass compilation?

```
class SuperMan{
    private int a;
    protected SuperMan(int a){this.a = a;}
}
...
class SubMan extends SuperMan{
    public SubMan(int a){super(a);}
    public SubMan(){super(5);}
}
```

HW5:

In C++, Kevin wrote the following code first

```
Class A {  
    A() ;  
    A(int a): foo(a){}  
    protected:  
        int foo ;  
};  
Class B : A {  
    B() ;  
    B(int a): A(a){}  
    protected:  
        string bar ;  
};
```

In main(), Kevin feel the need to initialize foo when needed

That is, he needs to declare a variable like

```
B b(100); // where 100 is to initialize foo.
```

Please modify the code the above to help Kevin.

HW6:

What is the program output of the following C++ code:

```
class A {
public:
    A();
    void foo() { cout << "A::foo"<< endl; }
    int bar() { cout << "A::bar" << endl ; }
};

class B : A {
public:
    B();
    void foo() { cout << "B::foo" << endl; }
    char bar() { cout << "B::bar" << endl ; }
}

main() {
    B b ;
    b.foo() ;
    b.bar();
}
```

Ans:

B::foo

B::bar